Yufeng Gu

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Education

University of Michigan **Ann Arbor, MI, USA**
Ph.D., Computer Science and Engineering Department *Sep. 2020 - Apr. 2026*

Zhejiang University (ZJU) **Hangzhou, China**
B.Eng., College of Information Science and Electronic Engineering *Sep. 2016 - June 2020*

Experience

Meta Platform Inc. **Menlo Park, CA, USA**
Research Scientist, AI system Co-design *May 2026 – Now*

Conduct research on architecting next-generator in-house AI datacenter. Co-design KV cache management policies and multi-tier memory and storage architectures for agentic AI workloads.

University of Michigan **Ann Arbor, MI, USA**
Graduate Research Assistant, Advisor: Reetuparna Das *Sep. 2020 – Apr. 2026*

Co-design architecture and system for large-scale emerging applications, including Generative AI and genome sequencing. Publish papers at top conferences including ISCA, ASPLOS and HPCA, awarded as Communication of ACM Research Highlights (24/10,000+) and HPCA Distinguished Artifact Honorable Mention (3/29).

Tenstorrent Inc. **Santa Clara, CA, USA**
Performance Architect Intern, Manager: Wei-han Lien *May 2023 – Aug. 2023*

Build a profile-guided agile performance model for Tensix AI accelerator, including computation and communication between RISC-V cores, with <10% error rate. This model reduces the performance modeling time from a few hours to ∼5 minutes, enabling fast kernel development and optimization.

Intel Labs **Hillsboro, OR, USA**
Graduate Research Intern, Manager: Nilesh Jain *June 2022 – Aug. 2022*

Develop an offline profile and online prediction-based resource scheduler for Quality of Service (QoS)-aware job co-location for ML inference, achieving up to 1.4× request throughput improvement over single model deployment and reducing scheduling times from ∼30 minutes to ∼10 seconds compared to state-of-the-art schedulers.

Yale University **New Heaven, CT, USA**
Undergraduate Research Intern, Advisor: James Duncan, Xiaoxiao Li *Nov. 2019 – Mar. 2020*

Interpretation on ASD with fMRI and Deep Learning Models. Publications on MICCAI and MedIA.

École Polytechnique Fédérale De Lausanne (EPFL) **Lausanne, Switzerland**
Undergraduate Research Intern, Advisor: Babak Falsafi *July 2019 – Sep. 2019*

Patched QFlex Simulator (from Makefile to CMake compilation) and accelerated activation functions for DNN.

Award & Honors

Outstanding Young Speaker Award (APPT 2025) *July 2025*
Hardware-Software Co-Design for Scalable LLM Inference

Distinguished Artifact Honorable Mention (HPCA 2025) *March 2025*
Multi-Dimensional Vector ISA Extension for Mobile In-Cache Computing. (3/29 Artifacts)

Communication of ACM (CACM) Research Highlights *July 2024*
GenDP: A Framework of Dynamic Programming Acceleration for Genome Sequencing Analysis.
CACM Research Highlights section selects 24/10,000+ papers from ACM conferences per year, reprinting the most significant and influential results across Computer Science.

University of Michigan Rackham Graduate Student Research Grant ($3000, role: PI) *Apr. 2024*
Pangenomics Benchmark Suite and Characterization.

University of Michigan Rackham Travel Grant, ISCA/HPCA Travel Grant *2023, 2025*

Fellowship of Summer@EPFL (2% applicants awarded) *July 2019*

Tang Lixin Fellowship (2‰ students awarded) *Nov. 2017, 2018, 2019*

Outstanding Student Leaders in Zhejiang University (3% students awarded) *Oct. 2017, 2019*

First-Class Scholarship for Outstanding Students (2% students awarded) *Oct. 2017*

Under-Review and Preprint Publications

Yufeng Gu, Jiazhen Wang, Reetuparna Das. "Balancing Generality and Specialization: A Survey on AI Datacenter Hardware Architecture." arXiv:2609.26829, 2026. *(Under Submission to ACM Computing Surveys)*

Sumanth Umesh, Ning Liang, **Yufeng Gu**, Reetuparna Das. "Addressing In-Memory Database Bottlenecks with CXL Memory Expansion." *(Under Submission to VLDB)*

Yufeng Gu, Reetuparna Das. "ADAPT: Context-Aware Scheduling for LLM Inference on a Heterogeneous Decoding Cluster" *(Under Submission to ISCA'27)*

Yufeng Gu, Samel Gobriel, Nilesh Jain, Reetuparna Das. "BRIDGE: A Systolic Processing-In-Memory Architecture for Narrow GEMMs in LLM Inference" *(Under Submission to IEEE Transactions on Computers)*

Jiazhen Wang, Finn Moore, **Yufeng Gu**, Shashwat Jaguri, Dong Eun Kim, Nicoletta Risi, Reetuparna Das. "SpikeGate: An Event-Driven Neuromorphic System for Efficient Edge Vision" *(Under Submission to DAC'27)*

Peer-Reviewed Publications

* equal contribution

[**12**] Pyeongsu Park, Hwijoon Lim, Seah Kim, Mustafa Ozdal, Yi-Chien Lin, Kevin Siu, Pavan Yalamanchili, Benjamin Ghaemmaghami, **Yufeng Gu**, Jongsoo Park, Joel Coburn, Paul Baumstarck, Shobhit Kanaujia, Chunqiang Tang. "AIPerf: Meta's Performance Modeling and HW/SW Co-design Framework for Datacenter-Scale LLM Training." In 59th IEEE/ACM International Symposium on Microarchitecture (MICRO'26), Industry Track.

[**11**] Danyang Chen*, **Yufeng Gu***, Yibo Huang, Chengxuan Pei, Peichun Hua, Yang Zhou, Yunming Xiao. "Unlocking Software-defined GPU Fabric Scheduling in the LLM Era." To appear in ACM SIGOPS Asia-Pacific Workshop on Systems (APSys'26).

[**10**] Jiajun Luo, Yu Jin, **Yufeng Gu**, Jinyi Deng, Yang Hu and Shuwen Deng. "Stone-Skimming: Attacking Big-Data Analytics Applications with Page Table Walk." In 63rd ACM/IEEE Design Automation Conference (DAC'26).

[**9**] Noah Kaplan, Jan-Niklas Schmelzle, **Yufeng Gu**, Christopher Batten, Reetuparna Das. "Pangenomics-Bench: A Benchmark Suite and Characterization of Pangenomics." In IEEE International Symposium on Workload Characterization (IISWC'25).

[**8**] **Yufeng Gu**, Arun Subramaniyan, Tim Dunn, Alireza Khadem, Kuan-Yu Chen, Somnath Paul, Md Vasimuddin, Sanchit Misra, David Blaauw, Satish Narayanasamy, Reetuparna Das. "GenDP: A Framework of Dynamic Programming Acceleration for Genome Sequencing Analysis." Communications of the ACM, 2025.

[7] Alireza Khadem*, Kamalavasan Kamalakkannan*, Zhenyan Zhu, Akash Poptani, **Yufeng Gu**, Jered Benjamin Dominguez-Trujillo, Nishil Talati, Daichi Fujiki, Scott Mahlke, Galen Shipman, Reetuparna Das. "DX100: Programmable Data Access Accelerator for Indirection." In Proceedings of the 52th Annual International Symposium on Computer Architecture (ISCA'25).

[6] **Yufeng Gu***, Alireza Khadem*, Sumanth Umesh, Ning Liang, Xavier Servot, Onur Mutlu, Ravishankar Iyer, Reetuparna Das. "PIM Is All You Need: A CXL-Enabled GPU-Free System for Large Language Model Inference." In Proceedings of the 30th ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS'25). **In Progress of Technology Transfer to SK Hynix's Next Generation Datacenter**

[5] Alireza Khadem, Daichi Fujiki, Hilbert Chen, **Yufeng Gu**, Nishil Talati, Scott Mahlke, Reetuparna Das. "Multi-Dimensional Vector ISA Extension for Mobile In-Cache Computing." In 2025 IEEE International Symposium on High-Performance Computer Architecture (HPCA'25). **Distinguished Artifact Honorable Mention (3/29)**

[4] **Yufeng Gu**, Arun Subramaniyan, Tim Dunn, Alireza Khadem, Kuan-Yu Chen, Somnath Paul, Md Vasimuddin, Sanchit Misra, David Blaauw, Satish Narayanasamy, Reetuparna Das. "GenDP: A Framework of Dynamic Programming Acceleration for Genome Sequencing Analysis." In Proceedings of the 50th Annual International Symposium on Computer Architecture (ISCA'23). **Communication of ACM Research Highlights (24/10,000+)**

[3] Arun Subramaniyan, **Yufeng Gu**, Timothy Dunn, Somnath Paul, Md Vasimuddin, Sanchit Misra, David Blaauw, Satish Narayanasamy, and Reetuparna Das. "GenomicsBench: A Benchmark Suite for Genomics." In IEEE International Symposium on Performance Analysis of Systems and Software (ISPASS'21).

[2] Xiaoxiao Li, **Yufeng Gu**, Nicha Dvornek, Lawrence H. Staib, Pamela Ventola, and James S. Duncan. "Multi-site fMRI analysis using privacy-preserving federated learning and domain adaptation: ABIDE results." Medical Image Analysis 65: 101765. *IF = 11.1*

[1] Xiaoxiao Li, Yuan Zhou, Nicha C. Dvornek, **Yufeng Gu**, Pamela Ventola, and James S. Duncan. "Efficient Shapley Explanation for Features Importance Estimation Under Uncertainty." In International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI'20).

Patents

[2] Reetuparna Das, **Yufeng Gu**, Alireza Khadem, Sumanth Unmesh. "Systolic Processing-In-Memory Architecture For Large Language Inference." PCT Application No. PCT/US2026/047307. Filed September 25, 2026.

[1] Reetuparna Das, **Yufeng Gu**, Alireza Khadem, Sumanth Unmesh. "Data Reuse-Aware Scalable Processing-In-Memory Architecture For Efficient Large Language Model Inference." U.S. Patent Application No. 19/243881. Filed August 28, 2025.

Talks

Architecture and System Co-Design for Scalable LLM Inference
Chinese University of Hong Kong (CUHK) *Dec. 2025*
PhD Forum at APPT 2025 *July 2025*
Los Alamos National Laboratory *May 2025*

CXL-enabled PIM system for LLM inference
SAFARI Live Seminar (Host: Prof. Onur Mutlu) *June 2025*
MCCSys workshop co-located with ASPLOS 2025 *Mar. 2025*

Genomics Benchmark Suite and Accelerator Design
Computer Architecture Seminar at UCF (Host: Prof. Di Wu) *Mar. 2024*
Cornel University (Host: Prof. Christopher Batten) *Feb. 2024*
Peisu Xia Forum at ICT, CAS *Dec. 2023*

Teaching

Guest Lecture at UMich EECS 570 Parallel Computer Architecture (Fall 2025 and Winter 2026)

Programming at Discovery Engineering program at University of Michigan *July. 2022*

Mathematics at Tuanlin Primary School, Guizhou, China *Aug. 2017*

Services

Program Committee
- IEEE International Symposium on High-Performance Computer Architecture (HPCA) 2027

Journal Reviewer
- IEEE Transaction on Computer (TC)
- Journal of Systems Architecture (JSA)
- IEEE Transaction on Big Data (TBD)
- IEEE Computer Architecture Letter (CAL)

Artifact Evaluation Committee
- International Symposium on Computer Architecture (ISCA) 2023, 2024, 2025
- IEEE International Symposium on Workload Characterization (IISWC) 2021

Mentorship

Linus Wang B.S. UM → Ph.D. at UMich *Jan. 2025 - Now*
Wenjie Geng M.S. UM → Research Assistant at UMich *Sep. 2024 - Aug. 2026*
Zesen Zhao M.S. UM → Ph.D. at UMich *Jan. 2025 - Aug. 2025*
Mayne Mei B.Eng. UM → Engineer at Etched *Jan. 2024 - Apr. 2024*
Yuzhe Ruan B.S. UM → M.S. at Yale *Sep. 2023 - Apr. 2024*
Donglin Yu B.Eng. → M.Eng. at UIUC *Jan. 2023 - Apr. 2024*
Dawit Melka B.Eng. Addis Ababa University → Engineer at iCog Labs *May 2022 - Aug. 2022*
James Gu B.Eng. UM → Engineer at Amazon *May 2022 - Aug. 2022*